

## Lecture 18

# Free fermions: the transverse-field Ising chain

**Objectives.** Course map: P5 begins; the model of G4. *Here is a chain of spins that looks interacting and is not. A change of variables due to Jordan and Wigner turns its spins into fermions and the Hamiltonian into a quadratic form; Fourier and Bogoliubov then diagonalise it completely. The lecture does this by hand, including the one step that short accounts drop: the chain splits into two sectors with different boundary conditions. The result is the model in which [Lecture 19](#) proves relaxation to a generalised Gibbs ensemble.*

## 18.1 The question

The chains and gases of [Lecture 10](#), [Lecture 11](#) and [Lecture 15](#)–[Lecture 17](#) needed the Bethe ansatz because their particles scatter. Some chains only look as if they do. So the question is

*which interacting-looking chains are secretly free, and how do you solve them completely?*

The answer, for the transverse-field Ising chain: the Jordan–Wigner map ([Theorem 18.3](#)) makes it quadratic in fermions, sector by sector, and Fourier and Bogoliubov give its whole spectrum in both parity sectors ([Theorem 18.6](#)). The worked example is the chain itself, against exact diagonalisation ([Figure 18.3](#)).

## 18.2 The models

On a ring of  $N$  sites, each carrying a spin  $\frac{1}{2}$  with Pauli matrices  $\sigma^x, \sigma^y, \sigma^z$ , the *transverse-field Ising model* (TFIM) is

$$H = -J \sum_{n=1}^N \sigma_n^x \sigma_{n+1}^x - h \sum_{n=1}^N \sigma_n^z, \quad J, h \geq 0, \quad \sigma_{N+1} = \sigma_1. \quad (18.1)$$

The first term wants neighbouring spins aligned along  $x$ ; the second, the *transverse* field, wants every spin up along  $z$ . The two do not commute, and that competition is the whole model.  $N$  is **even** throughout. (For odd  $N$  the momentum  $k = \pi$  moves from one of the grids of § 18.5 to the other; nothing changes in principle, but the bookkeeping does, and it is not treated.)

*Two limits by inspection.* At  $J = 0$  the ground state is  $|\uparrow\uparrow \cdots \uparrow\rangle$  with energy  $-hN$ , and flipping one spin costs  $2h$ . At  $h = 0$  there are two ground states, all spins along  $+x$  or all along  $-x$ , energy  $-JN$ ; the cheapest excitation of a ring is a pair of *domain walls*, each

breaking one bond at cost  $2J$ , so  $4J$  in all. Between the limits something must happen, and the reading at the end finds it at  $h = J$ .

*The symmetry.* Let

$$\Pi = \prod_{n=1}^N \sigma_n^z. \quad (18.2)$$

Each  $\sigma_n^z$  commutes with the field term and anticommutes with  $\sigma_n^x$ , so  $\Pi$  commutes with every bond  $\sigma_n^x \sigma_{n+1}^x$  (two anticommutations) and with  $H$ . It is the spin flip  $\sigma^x \rightarrow -\sigma^x$ , it squares to 1, and it splits the Hilbert space into an *even* sector  $\Pi = +1$  and an *odd* sector  $\Pi = -1$ . It exchanges the two  $h = 0$  ground states, so the combinations with a definite parity are their sum (even) and difference (odd).

## 18.3 Fermions

### The algebra

**Definition 18.1 (Fermionic modes).** Operators  $c_1, \dots, c_N$  on a Hilbert space satisfy the *canonical anticommutation relations* (CAR) if

$$\{c_m, c_n^\dagger\} = \delta_{mn}, \quad \{c_m, c_n\} = 0, \quad (18.3)$$

where  $\{A, B\} = AB + BA$ . The operator  $n_j = c_j^\dagger c_j$  is the *occupation* of mode  $j$ , and  $N_f = \sum_j n_j$  the *fermion number*.

Everything about these operators follows from (18.3). Taking  $m = n$  in the second relation gives  $c_j^2 = 0$  and  $(c_j^\dagger)^2 = 0$ : no mode can be created twice — Pauli's principle. Then  $n_j^2 = c_j^\dagger(1 - c_j^\dagger c_j)c_j = n_j$ , so each  $n_j$  has eigenvalues 0 and 1, and the  $n_j$  commute with each other.

*Fock space.* Suppose a unit vector  $|0\rangle$ , the *vacuum*, has  $c_j|0\rangle = 0$  for all  $j$ . The  $2^N$  vectors

$$c_{j_1}^\dagger c_{j_2}^\dagger \cdots c_{j_M}^\dagger |0\rangle, \quad j_1 < j_2 < \cdots < j_M,$$

are eigenvectors of every  $n_j$ , with  $n_j = 1$  exactly for the  $j$  listed; they are orthonormal, their span is mapped to itself by every  $c_j$  and  $c_j^\dagger$ , and the action there is fixed by the CAR alone. The order of the  $j$ 's matters only through a sign:  $c_1^\dagger c_2^\dagger |0\rangle = -c_2^\dagger c_1^\dagger |0\rangle$ . That sign is what makes fermions fermions, and it is where spins differ.

### Spins are not fermions

On one site,  $\sigma^+ = |\uparrow\rangle\langle\downarrow|$  and  $\sigma^- = |\downarrow\rangle\langle\uparrow|$  satisfy  $\{\sigma^-, \sigma^+\} = 1$  and  $(\sigma^-)^2 = 0$ : one spin is one fermionic mode, with  $|\uparrow\rangle$  empty and  $|\downarrow\rangle$  occupied. But on two different sites,  $\sigma_1^-$  and  $\sigma_2^-$  act on different tensor factors, so they *commute*; their anticommutator is  $2\sigma_1^- \sigma_2^- \neq 0$ . Spin flips on different sites are hard-core bosons: they exclude each other on a site and commute between sites. The fix has to supply a minus sign whenever two flips are exchanged, and it cannot be local.

## 18.4 The Jordan–Wigner map

**Definition 18.2** (Jordan–Wigner). On  $N$  spins, set

$$c_n = \left( \prod_{m < n} \sigma_m^z \right) \sigma_n^+, \quad c_n^\dagger = \left( \prod_{m < n} \sigma_m^z \right) \sigma_n^-. \quad (18.4)$$

The product  $S_n = \prod_{m < n} \sigma_m^z$  is the *string*;  $S_1 = 1$ .



Figure 18.1: The Jordan–Wigner string. Creating a fermion at site  $n$  flips the spin there down and multiplies by  $\pm 1$  according to the parity of the fermions (down spins) to its left. The sign is non-local, and it is exactly the sign an exchange of two fermions requires.

A fermion is a *down* spin, and the string counts how many fermions stand to the left (Figure 18.1): since  $\sigma^z = 1$  on  $|\uparrow\rangle$  and  $-1$  on  $|\downarrow\rangle$ ,

$$n_n = c_n^\dagger c_n = \sigma_n^- \sigma_n^+ = \frac{1}{2}(1 - \sigma_n^z), \quad S_n = \prod_{m < n} (1 - 2n_m), \quad \Pi = (-1)^{N_f}. \quad (18.5)$$

The strings cancel in  $n_n$  because  $S_n^2 = 1$  and  $S_n$  acts on other sites than  $\sigma_n^\pm$ . The vacuum is  $|\uparrow \cdots \uparrow\rangle$ , and the spin parity (18.2) is the fermion parity.

**Theorem 18.3** (Jordan–Wigner preserves the algebra). *The operators (18.4) satisfy the CAR (18.3). Conversely  $\sigma_n^+ = S_n c_n$  and  $\sigma_n^- = 1 - 2c_n^\dagger c_n$ , so every spin operator is a function of the  $c$ 's.*

*Proof.* Everything rests on one fact about a single spin:

$$\sigma^z \sigma^\pm = -\sigma^\pm \sigma^z \quad (18.6)$$

( $\sigma^z \sigma^+ = \sigma^+$ , since  $\sigma^+$  outputs  $|\uparrow\rangle$ ; and  $\sigma^+ \sigma^z = -\sigma^+$ , since it accepts only  $|\downarrow\rangle$ ). *Same site.*  $S_n$  commutes with  $\sigma_n^\pm$  and  $S_n^2 = 1$ , so  $\{c_n, c_n^\dagger\} = S_n \{\sigma_n^+, \sigma_n^-\} S_n = 1$  and  $c_n^2 = S_n^2 (\sigma_n^+)^2 = 0$ . *Different sites,  $m < n$ .* Write  $S_n = S_m \sigma_m^z T$  with  $T = \prod_{m < l < n} \sigma_l^z$ . The operators  $S_m, T, \sigma_n^\pm$  act on sites other than  $m$  and commute with everything at site  $m$ ; so

$$c_m c_n = S_m \sigma_m^+ S_m \sigma_m^z T \sigma_n^+ = \sigma_m^+ \sigma_m^z T \sigma_n^+, \quad c_n c_m = S_m \sigma_m^z T \sigma_n^+ S_m \sigma_m^+ = \sigma_m^z \sigma_m^+ T \sigma_n^+,$$

and (18.6) gives  $c_m c_n = -c_n c_m$ . Replacing  $\sigma_n^+$  by  $\sigma_n^-$ , or  $\sigma_m^+$  by  $\sigma_m^-$ , changes nothing in the argument, which gives  $\{c_m, c_n^\dagger\} = 0$ ;  $m > n$  is the same with the names exchanged, and the remaining relations are adjoints. The converse is (18.4) multiplied by  $S_n$ , and (18.5). ■

### The Hamiltonian in fermions

The field term is immediate:  $-h\sigma_n^z = 2hn_n - h$ . A bond between neighbours is almost as easy, because the strings of  $n$  and  $n+1$  differ by one factor.

**Proposition 18.4** (Bonds and the boundary term). *With  $c$ 's as in (18.4),*

$$\sigma_n^x \sigma_{n+1}^x = (c_n^\dagger - c_n)(c_{n+1}^\dagger + c_{n+1}), \quad 1 \leq n \leq N-1, \quad (18.7)$$

$$\sigma_N^x \sigma_1^x = -\Pi (c_N^\dagger - c_N)(c_1^\dagger + c_1). \quad (18.8)$$

*Proof.* By Theorem 18.3,  $\sigma_n^x = \sigma_n^+ + \sigma_n^- = S_n(c_n + c_n^\dagger)$ . First, the one-site identities  $(1-2n)c = c$  and  $(1-2n)c^\dagger = -c^\dagger$  (because  $nc = 0$  and  $nc^\dagger = c^\dagger$ ) give  $(c + c^\dagger)(1-2n) = c^\dagger - c$  and  $(1-2n)(c + c^\dagger) = c - c^\dagger$ . *Bulk.*  $S_{n+1} = S_n \sigma_n^z$ , and  $S_n$  commutes with  $c_n$ , so

$$\sigma_n^x \sigma_{n+1}^x = S_n(c_n + c_n^\dagger) S_n \sigma_n^z (c_{n+1} + c_{n+1}^\dagger) = (c_n + c_n^\dagger)(1-2n_n)(c_{n+1} + c_{n+1}^\dagger),$$

which is (18.7). *Boundary.* Now the string is the whole chain but site  $N$ :  $S_N = \Pi \sigma_N^z$ , and  $S_1 = 1$ . So  $\sigma_N^x \sigma_1^x = \Pi (1-2n_N)(c_N + c_N^\dagger)(c_1 + c_1^\dagger) = \Pi (c_N - c_N^\dagger)(c_1 + c_1^\dagger)$ . ■

The boundary bond is the one term that is *not* a quadratic form in the  $c$ 's: it carries the parity of the whole chain. But  $\Pi$  commutes with  $H$ , so on each sector it is a number, and there the boundary term is quadratic after all. Define  $c_{N+1} := -c_1$  in the even sector and  $c_{N+1} := +c_1$  in the odd sector. Then in both, (18.8) reads  $(c_N^\dagger - c_N)(c_{N+1}^\dagger + c_{N+1})$ , the same form as the bulk, and

$$H = H_+ \frac{1+\Pi}{2} + H_- \frac{1-\Pi}{2}, \quad H_\pm = -J \sum_{n=1}^N (c_n^\dagger - c_n)(c_{n+1}^\dagger + c_{n+1}) + 2h \sum_{n=1}^N c_n^\dagger c_n - hN, \quad (18.9)$$

with  $c_{N+1} = \mp c_1$ : the fermions are *antiperiodic in the even sector and periodic in the odd one*. Each  $H_\pm$  is quadratic, commutes with  $\Pi$  (every term changes  $N_f$  by 0 or  $\pm 2$ ), and acts on all of Fock space. So the spectrum of  $H$  is found in two steps: diagonalise  $H_+$  and  $H_-$  completely, then keep the even states of the first and the odd states of the second. The half thrown away is not part of the spin chain.

## 18.5 Fourier and Bogoliubov

### Two grids of momenta

In each sector the fermions obey  $c_{n+N} = \mp c_n$ , so the plane waves that respect this are  $e^{ikn}$  with  $e^{ikN} = \mp 1$ :

$$\text{even sector (NS): } k \in \left\{ \frac{\pi(2m+1)}{N} \right\}, \quad \text{odd sector (R): } k \in \left\{ \frac{2\pi m}{N} \right\}, \quad m = 0, \dots, N-1, \quad (18.10)$$

with  $k$  taken mod  $2\pi$ . (The names are borrowed from string theory, Neveu–Schwarz and Ramond, where the same two boundary conditions occur.) Both grids are symmetric under  $k \rightarrow -k$ . The R grid contains  $k = 0$  and  $k = \pi$ , which are their own negatives; the NS grid, for even  $N$ , contains neither (Figure 18.2). In the sector's grid set

$$c_n = \frac{1}{\sqrt{N}} \sum_k e^{ikn} c_k, \quad c_k = \frac{1}{\sqrt{N}} \sum_{n=1}^N e^{-ikn} c_n. \quad (18.11)$$

The matrix  $(e^{-ikn}/\sqrt{N})$  is unitary, so the  $c_k$  satisfy the CAR, and  $c_{N+1}$  computed from (18.11) is  $\mp c_1$  automatically.

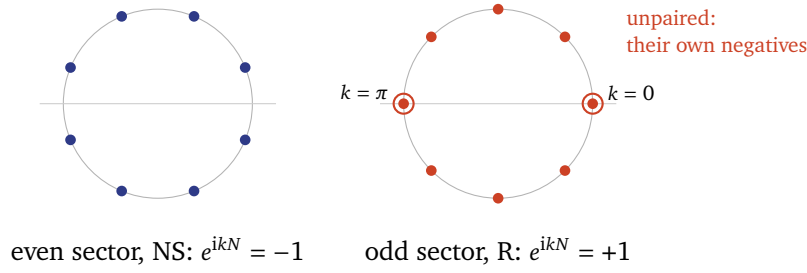


Figure 18.2: The two grids of momenta for  $N = 8$ , drawn as  $e^{ik}$  on the unit circle. Each point is paired with its mirror image  $-k$ . The R grid has two points on the real axis,  $k = 0$  and  $k = \pi$ , that pair with nothing; the NS grid has none.

**Proposition 18.5** (The Hamiltonian in momentum space). *In each sector, with  $k$  running over that sector's grid,*

$$H_{\pm} = \sum_k \left[ 2(h - J \cos k) c_k^{\dagger} c_k - iJ \sin k (c_k^{\dagger} c_{-k}^{\dagger} - c_{-k} c_k) \right] - hN. \quad (18.12)$$

*Proof.* Substitute (18.11) and use  $\frac{1}{N} \sum_n e^{i(q-k)n} = \delta_{kq}$  on the grid. The field gives  $\sum_n c_n^{\dagger} c_n = \sum_k c_k^{\dagger} c_k$ . Expanding the bond,  $(c_n^{\dagger} - c_n)(c_{n+1}^{\dagger} + c_{n+1}) = c_n^{\dagger} c_{n+1} + c_{n+1}^{\dagger} c_n + c_n^{\dagger} c_{n+1}^{\dagger} + c_{n+1} c_n$ . *Hopping:*  $\sum_n c_n^{\dagger} c_{n+1} = \sum_k e^{ik} c_k^{\dagger} c_k$ , and with its conjugate this gives  $2 \cos k$ . *Pairing:*  $\sum_n c_n^{\dagger} c_{n+1}^{\dagger} = \sum_k e^{ik} c_k^{\dagger} c_{-k}^{\dagger}$ ; relabelling  $k \rightarrow -k$  and using  $c_{-k}^{\dagger} c_k^{\dagger} = -c_k^{\dagger} c_{-k}^{\dagger}$  turns this into  $i \sum_k \sin k c_k^{\dagger} c_{-k}^{\dagger}$ , and the last term is its conjugate. ■

The pairing term creates or destroys fermions two at a time, at  $k$  and  $-k$ : the fermion number is not conserved, only its parity. So group the modes in pairs. For  $0 < k < \pi$  on the grid write  $\Psi_k = (c_k, c_{-k}^{\dagger})^T$ ,

$$a_k = 2(h - J \cos k), \quad b_k = 2J \sin k, \quad h_k = \begin{pmatrix} a_k & -ib_k \\ ib_k & -a_k \end{pmatrix}.$$

Collecting the terms of  $k$  and  $-k$  in (18.12), and reordering  $c_{-k}^{\dagger} c_{-k} = 1 - c_{-k} c_{-k}^{\dagger}$ ,

$$H_{\pm} = \sum_{0 < k < \pi} \Psi_k^{\dagger} h_k \Psi_k + \sum_{k \in \{0, \pi\}} a_k (c_k^{\dagger} c_k - \frac{1}{2}), \quad (18.13)$$

where the second sum is present only in the odd sector. The constants cancel: the reordering produces  $\sum_{0 < k < \pi} a_k$ , and  $\sum_k \cos k = 0$  on either grid, so  $\sum_{0 < k < \pi} a_k - hN = -\frac{1}{2} \sum_{k \in \{0, \pi\}} a_k$ .

### Bogoliubov

A  $2 \times 2$  Hermitian matrix is diagonalised by a rotation, and  $h_k$  has eigenvalues  $\pm \sqrt{a_k^2 + b_k^2} = \pm \varepsilon(k)$ , where

$$\varepsilon(k) = 2\sqrt{J^2 + h^2 - 2Jh \cos k} \geq 0. \quad (18.14)$$

Let  $\theta_k$  be the angle with  $\cos 2\theta_k = a_k/\varepsilon(k)$ ,  $\sin 2\theta_k = b_k/\varepsilon(k)$  (for  $J = h = 0$  the whole block vanishes and any  $\theta_k$  will do; otherwise  $\varepsilon(k) > 0$  at every paired  $k$ , since it vanishes only at  $k = 0$  when  $h = J$ ), and

$$\gamma_k = \cos \theta_k c_k - i \sin \theta_k c_{-k}^{\dagger}, \quad \gamma_{-k} = \cos \theta_k c_{-k} + i \sin \theta_k c_k^{\dagger}. \quad (18.15)$$

In matrix form  $\Psi_k = U_k \Gamma_k$  with  $\Gamma_k = (\gamma_k, \gamma_{-k}^\dagger)^T$  and  $U_k = \begin{pmatrix} \cos \theta_k & i \sin \theta_k \\ i \sin \theta_k & \cos \theta_k \end{pmatrix}$ , and a two-line computation gives

$$U_k^\dagger h_k U_k = \begin{pmatrix} a_k \cos 2\theta_k + b_k \sin 2\theta_k & i(a_k \sin 2\theta_k - b_k \cos 2\theta_k) \\ -i(a_k \sin 2\theta_k - b_k \cos 2\theta_k) & -(a_k \cos 2\theta_k + b_k \sin 2\theta_k) \end{pmatrix} = \begin{pmatrix} \varepsilon(k) & 0 \\ 0 & -\varepsilon(k) \end{pmatrix}.$$

Three things make (18.15) a legitimate change of fermions. The  $\gamma$ 's satisfy the CAR —  $\{\gamma_k, \gamma_k^\dagger\} = \cos^2 + \sin^2 = 1$ ,  $\{\gamma_k, \gamma_{-k}\} = i \sin \theta_k \cos \theta_k - i \sin \theta_k \cos \theta_k = 0$ , and the rest are immediate. The formula for  $\gamma_{-k}$  is the formula for  $\gamma_k$  at  $-k$ , since  $\theta_{-k} = -\theta_k$  ( $a$  is even in  $k$ ,  $b$  odd). And the state  $|\Omega_k\rangle = (\cos \theta_k + i \sin \theta_k c_k^\dagger c_{-k}^\dagger) |0\rangle$  is killed by both  $\gamma_k$  and  $\gamma_{-k}$ , as one checks in a line: it is the pair's new vacuum, and it contains an even number of fermions. Then

$$\Psi_k^\dagger h_k \Psi_k = \varepsilon(k) (\gamma_k^\dagger \gamma_k - \gamma_{-k} \gamma_{-k}^\dagger) = \varepsilon(k) (\gamma_k^\dagger \gamma_k + \gamma_{-k}^\dagger \gamma_{-k} - 1).$$

**Theorem 18.6** (The spectrum of the transverse-field Ising chain). *Let  $N$  be even and  $J, h \geq 0$ . Every eigenvalue of (18.1) is one of*

$$\text{even sector: } E = \sum_{k \in \text{NS}} \varepsilon(k) (n_k - \tfrac{1}{2}), \quad \sum_k n_k \text{ even}, \quad (18.16)$$

$$\text{odd sector: } E = \sum_{k \in \text{R}} e_k (n_k - \tfrac{1}{2}), \quad \sum_k n_k \text{ odd}, \quad (18.17)$$

with each  $n_k \in \{0, 1\}$ , and every such choice gives exactly one eigenvector. In the odd sector  $e_k = \varepsilon(k)$  for  $k \neq 0, \pi$ , while the two unpaired modes carry the signed energies

$$e_0 = 2(h - J), \quad e_\pi = 2(h + J). \quad (18.18)$$

In particular the lowest even level is  $E_0^+ = -\frac{1}{2} \sum_{k \in \text{NS}} \varepsilon(k)$ .

*Proof.* In the even sector, (18.13) and the Bogoliubov step give  $H_+ = \sum_{k \in \text{NS}} \varepsilon(k) (\gamma_k^\dagger \gamma_k - \frac{1}{2})$ , a sum of commuting occupations of  $N$  fermionic modes, whose common eigenvectors are the  $2^N$  occupation states over the vacuum  $\prod_{0 < k < \pi} (\cos \theta_k + i \sin \theta_k c_k^\dagger c_{-k}^\dagger) |0\rangle$ . In the odd sector the same holds with  $N - 2$  Bogoliubov modes and the two unpaired  $c_0, c_\pi$ , whose terms in (18.13) are  $a_0 = 2(h - J)$  and  $a_\pi = 2(h + J)$  — not  $\varepsilon$ , because nothing pairs them and nothing makes their sign positive. It remains to see which states each sector keeps.  $\Pi$  anticommutes with every  $c_n$ , hence with every  $c_k$  and every  $\gamma_k$ ; the vacuum is even, being built from  $|0\rangle$  by the even operators that make the  $|\Omega_k\rangle$ ; so the occupation state  $\{n_k\}$  has  $\Pi = (-1)^{\sum n_k}$ . By (18.9) the even sector keeps the even occupations of  $H_+$ , and the odd sector the odd occupations of  $H_-$ . ■

*The board check.* Take  $h = 0$ . Then  $\varepsilon(k) = 2J$  for every  $k$ , and the even sector's lowest level is  $-\frac{1}{2} \cdot N \cdot 2J = -JN$ . In the odd sector  $e_0 = -2J$  is negative: occupying  $k = 0$  lowers the energy. Its lowest level has  $n_0 = 1$  and everything else empty — one fermion, odd, allowed — and equals  $-\frac{1}{2}(-2J + 2J + (N - 2)2J) - 2J = -JN$ . The two  $h = 0$  ground states of § 18.2 are reproduced, one in each sector. Had the odd sector been given NS momenta, or had  $e_0$  been replaced by  $|e_0|$ , its lowest level would have come out at  $-JN + 2J$  and the degeneracy would have been lost. Figure 18.3 compares the theorem with exact diagonalisation for  $N = 8$  at twenty-one fields.

Figure 18.4 shows  $\varepsilon(k)$  for three fields.

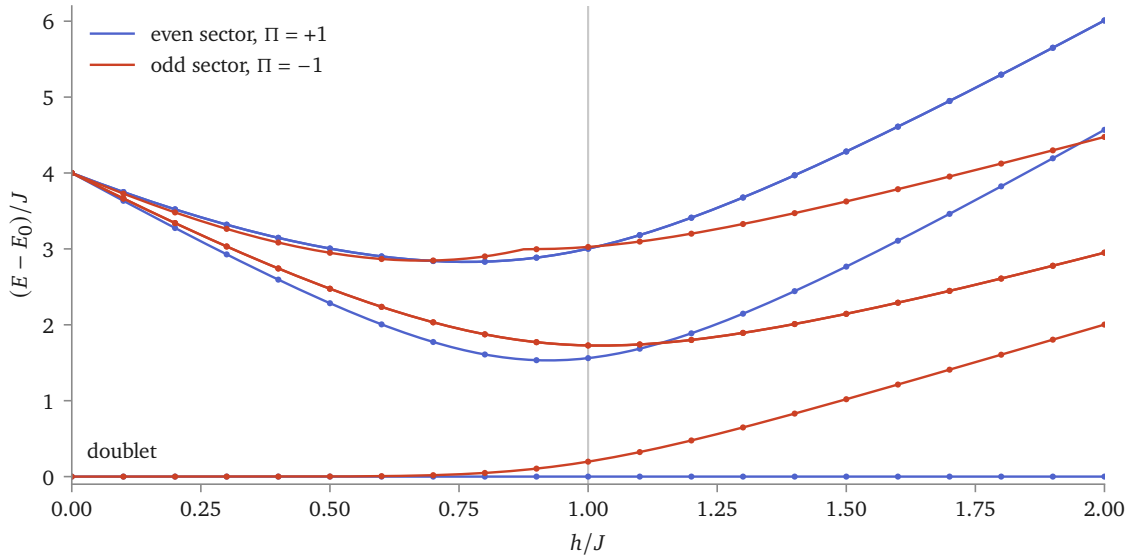


Figure 18.3: The lowest levels of each parity sector of the TFIM,  $N = 8$ , from the ground energy. Lines: [Theorem 18.6](#); dots: exact diagonalisation. The generator (`scripts/figures/tfim_spectrum.py`) asserts agreement of each sector's whole spectrum to  $10^{-9}$  at every dotted field. Levels at  $\pm k$  coincide.

## 18.6 What it buys

- *A complete solution:* every eigenvalue and eigenvector of the chain, in both parity sectors ([Theorem 18.6](#)).
- *A conserved occupation  $n_k$  for every mode,* the data [Lecture 19](#) needs to say what a relaxed state remembers.
- *The same free fermions elsewhere:* at infinite repulsion [Lecture 16](#)'s gas has the free-fermion ground-state energy ([Proposition 16.6](#)), and Girardeau's map makes it free fermions for every state, with a boundary condition that depends on  $N$  (the reading at the end).

## 18.7 Caveats, in one place

- *Status.* [Theorem 18.3](#) and [Theorem 18.6](#) are proved; the spectrum is also checked against exact diagonalisation at  $N = 8$ .
- *$N$  is even* throughout; odd  $N$  changes the bookkeeping of the grids and is not treated.
- *The two sectors are not optional.* One boundary condition for both, or none, gives a quadratic model that is simple and wrong.
- *Free is special.* Jordan–Wigner turns the Heisenberg chain's  $\sigma^x \sigma^z$  term into a quartic one: its fermions interact, and it needed the Bethe ansatz.

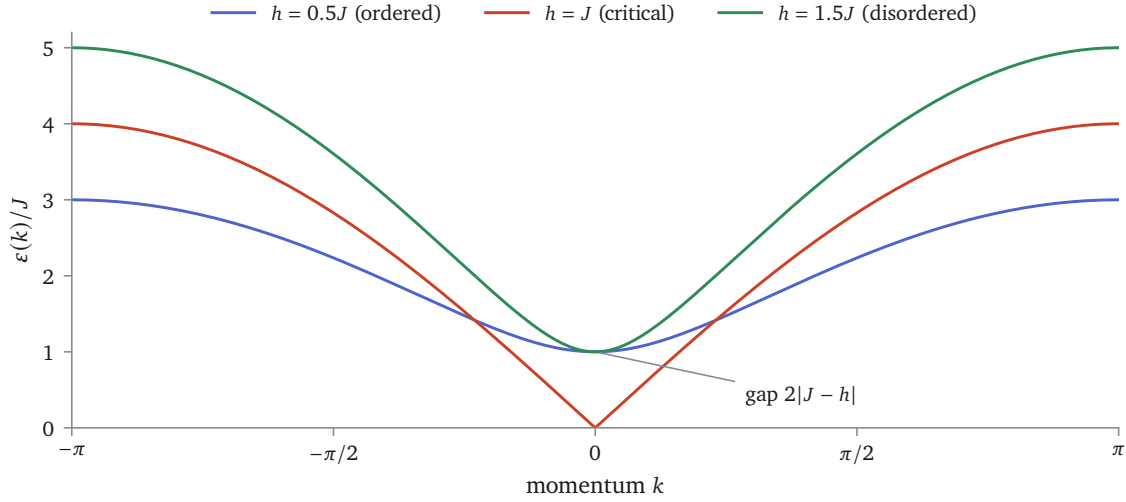


Figure 18.4: The quasiparticle energy (18.14) below, at and above the critical field. Its minimum, at  $k = 0$ , is  $2|J - h|$ , the same for  $h = 0.5J$  and  $h = 1.5J$ ; at  $h = J$  it closes linearly. (scripts/figures/tfim\_dispersion.py.)

Tutorial

## 18.8 Jordan–Wigner by hand, and local charges

*Three sites by hand.* On three sites,  $c_1 = \sigma_1^+$ ,  $c_2 = \sigma_1^z \sigma_2^+$ ,  $c_3 = \sigma_1^z \sigma_2^z \sigma_3^+$ . Then  $\{c_1, c_2\} = (\sigma_1^+ \sigma_2^z + \sigma_2^z \sigma_1^+) \sigma_2^+ = 0$ , where  $\sigma_1^+$  and  $\sigma_2^+$  alone commute. On states,

$$c_2^\dagger c_3^\dagger |\uparrow\uparrow\uparrow\rangle = \sigma_2^- \sigma_2^z \sigma_3^- |\uparrow\uparrow\uparrow\rangle = |\uparrow\downarrow\downarrow\rangle, \quad c_3^\dagger c_2^\dagger |\uparrow\uparrow\uparrow\rangle = \sigma_2^z \sigma_2^- \sigma_3^- |\uparrow\uparrow\uparrow\rangle = -|\uparrow\downarrow\downarrow\rangle:$$

the string at site 3 sees the fermion at site 2 or not, according to the order of creation.

*The construction of local charges.* Every occupation  $\gamma_k^\dagger \gamma_k$  commutes with  $H_\pm$ ; the local charges come from weighting whole blocks: for an even function  $f$ , weight the term of  $k$  in (18.13) by  $f(k)$ . The result commutes with  $H_\pm$  block by block, and if  $f(k) = \cos mk$  the weighted coefficients are still trigonometric polynomials in  $k$ , so the operator is a sum of fermion bilinears of range at most  $m+2$ . The odd combinations  $\sum_k \sin mk c_k^\dagger c_k$  also commute, because pairing creates  $k$  and  $-k$  together and leaves  $n_k - n_{-k}$  alone. In spin language the first few are these.

**Proposition 18.7** (Local charges of the TFIM · sketched). *Beside  $H$ , the local charges*

$$Q_+ = \sum_{n=1}^N \left[ J(\sigma_n^x \sigma_{n+1}^z \sigma_{n+2}^x - \sigma_n^z) - h(\sigma_n^x \sigma_{n+1}^x + \sigma_n^y \sigma_{n+1}^y) \right],$$

$$Q_- = \sum_{n=1}^N (\sigma_n^x \sigma_{n+1}^y - \sigma_n^y \sigma_{n+1}^x), \quad Q'_- = \sum_{n=1}^N (\sigma_n^x \sigma_{n+1}^z \sigma_{n+2}^y - \sigma_n^y \sigma_{n+1}^z \sigma_{n+2}^x)$$

*commute with  $H$  and with each other.  $Q_+$  is, in each sector,  $-2$  times the  $f = \cos k$  charge above, plus a constant. (The fermionic argument is the sketch; the commutators are checked exactly for  $N = 6, 8, 10$ .)*

At  $N = 8$  and  $h = 0.7J$  a computer search (the check script) finds that, modulo multiples of the identity, the conserved local charges of range at most three are spanned by  $H$ ,  $Q_+$ ,

$Q_-$ ,  $Q'_-$  — at that size and coupling only. Spin forms follow from (18.7) and its analogues: a fermion bilinear on sites  $n$  and  $n + d$  carries the string of the  $d - 1$  sites between them, and on the ring the terms that wrap round pick up  $-\Pi$  exactly as the boundary bond did, so one spin operator serves both sectors.

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Reading

## 18.9 The quantum phase transition

Gaps. From (18.14),

$$\varepsilon(k)^2 - 4(J - h)^2 = 8Jh(1 - \cos k) \geq 0, \quad (18.19)$$

so the cheapest quasiparticle, at  $k = 0$ , costs  $2|J - h|$ : the *quasiparticle gap*. It closes at  $h = J$  and nowhere else. The *many-body gap* — the distance from the ground state to the next level of the chain — is a different number, because each sector keeps only one parity of quasiparticle number. Riemann sums over the two grids are compared with one observation: for a  $2\pi$ -periodic function  $f = \sum_m \hat{f}_m e^{imk}$ ,  $\sum_n e^{imk_n}$  over either grid vanishes unless  $N$  divides  $m$ , so

$$\sum_{k \in \text{NS}} f(k) - \sum_{k \in \text{R}} f(k) = -2N \sum_{j \text{ odd}} \hat{f}_{jN}, \quad (18.20)$$

which tends to 0 as  $N \rightarrow \infty$  for smooth  $f$ , whose Fourier coefficients decay faster than any power.

**Theorem 18.8 (Gaps of the TFIM).** *As  $N \rightarrow \infty$  through even values:*

- (a) *if  $h > J$ , the ground state is the even sector's lowest state and the many-body gap tends to  $2(h - J)$ , the next level being the odd sector's lowest, one quasiparticle at  $k = 0$ ;*
- (b) *if  $h < J$ , the lowest even and odd levels become degenerate, and the gap above this doublet tends to  $4(J - h)$ ;*
- (c) *· sketched the doublet's splitting decays exponentially in  $N$ , roughly like  $(h/J)^N$ .*

*Proof.* (a) For  $h > J$  both signed energies (18.18) are positive and equal to  $\varepsilon(0)$ ,  $\varepsilon(\pi)$ . The even sector's excited states have at least two quasiparticles, so lie at least  $4(h - J)$  above  $E_0^+$ . The odd sector's lowest state has one quasiparticle, the cheapest, at  $k = 0$ :  $E_0^- = -\frac{1}{2} \sum_{\text{R}} \varepsilon + \varepsilon(0)$ , so  $E_0^- - E_0^+ = \varepsilon(0) + \frac{1}{2} (\sum_{\text{NS}} \varepsilon - \sum_{\text{R}} \varepsilon) \rightarrow 2(h - J)$  by (18.20);  $\varepsilon$  is smooth for  $h \neq J$ . (b) For  $h < J$ ,  $e_0 = -\varepsilon(0) < 0$ , so the odd sector's lowest state occupies  $k = 0$  and nothing else — odd, as required — and  $E_0^- = -\frac{1}{2} \sum_{\text{R}} |e_k| = -\frac{1}{2} \sum_{\text{R}} \varepsilon$ . Then  $E_0^- - E_0^+ \rightarrow 0$  by (18.20). Any other state of either sector differs from that sector's lowest state in an even number, at least two, of occupations, each changing the energy by some  $|e_k| \geq 2(J - h)$ ; so it lies at least  $4(J - h)$  above, and two quasiparticles at  $k = \pm\pi/N$  in the even sector come within  $o(1)$  of that bound. (c)  $\varepsilon$  extends analytically to complex  $k$  up to its branch points, where  $e^{ik} = h/J$  or  $J/h$ ; so  $\hat{e}_m$  decays like  $(h/J)^{|m|}$  and (18.20) is dominated by  $j = \pm 1$ . ■

For  $N = 8$  and  $h = 0.4J$  the numbers are: doublet splitting  $2.6 \cdot 10^{-4}J$ , against a quasiparticle gap of  $1.2J$ , and the next level  $2.60J$  above the ground state, on its way to  $4(J - h) = 2.4J$  (check-l14-algebra.py). Each extra two sites cut the splitting by a factor approaching  $(h/J)^2$ . At  $h = 0$ , the gap  $4J$  above the doublet is the two domain walls of § 18.2.

**Two phases.** The doublet is spontaneous symmetry breaking seen through a finite window. In the ordered phase  $h < J$  the even and odd ground states are the finite- $N$  versions of  $|\rightarrow\rightarrow\cdots\rangle \pm |\leftarrow\leftarrow\cdots\rangle$ ; their splitting is the amplitude to tunnel between the two ordered states, which needs a domain wall to sweep round the whole ring — hence exponentially small. As  $N \rightarrow \infty$  they become degenerate, and their combinations break the  $\Pi$  symmetry, with  $\langle \sigma^x \rangle \neq 0$  (the reading computes it). In the disordered phase  $h > J$  the ground state is unique and gapped, and a quasiparticle is a dressed flipped spin.

Between the phases the ground-state energy per site,

$$e_0(h) = \lim_{N \rightarrow \infty} \frac{E_0^+}{N} = -\frac{1}{2\pi} \int_0^\pi \varepsilon(k) dk, \quad (18.21)$$

is continuous and has a continuous first derivative, but its second derivative diverges at  $h = J$ , like  $\frac{1}{\pi J} \ln |h - J|$  (Figure 18.5) · *sketched*: near  $k = 0$  and  $h = J$ ,  $\varepsilon \approx 2\sqrt{(h - J)^2 + J^2 k^2}$ , whose second  $h$ -derivative integrates to a logarithm. A non-analyticity of the ground-state energy is what a *quantum phase transition* is, and the closing gap is its signature. At  $h = J$  itself,  $\varepsilon(k) = 4J |\sin(k/2)|$  and  $e_0 = -4J/\pi$ .

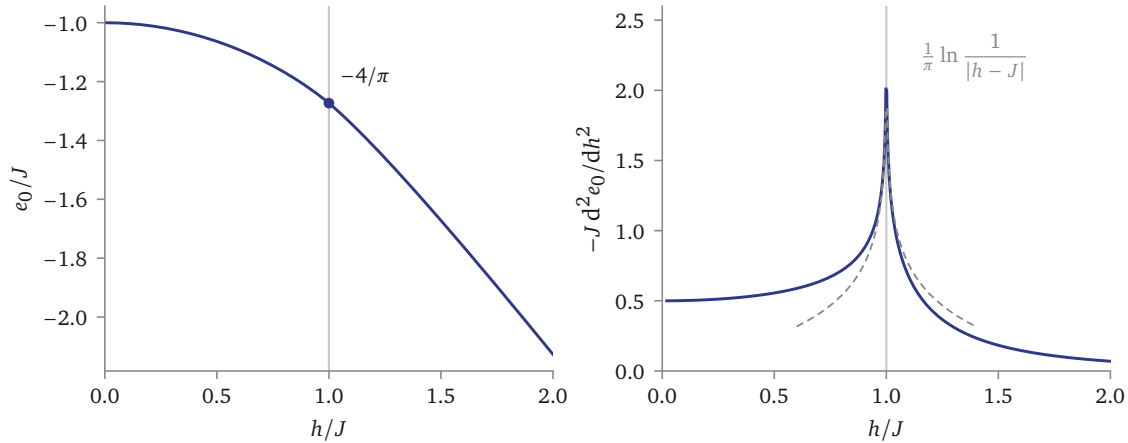


Figure 18.5: Left: the ground-state energy per site (18.21); it looks smooth. Right: minus its second derivative, which diverges at  $h = J$ , with the asymptote  $\frac{1}{\pi} \ln \frac{1}{|h-J|}$  (dashed,  $J = 1$ ). The generator asserts the slope  $1/\pi$  to 3% on both sides.

**Duality.**

**Proposition 18.9** (Kramers–Wannier, on the spectrum).  $\varepsilon(k)$  is symmetric under  $J \leftrightarrow h$ . Consequently the whole even-sector spectrum (18.16) at finite  $N$ , and  $e_0$ , are unchanged by  $J \leftrightarrow h$ ; the quasiparticle gap closes only at the self-dual point  $h = J$ .

*Proof.*  $J^2 + h^2 - 2Jh \cos k$  is symmetric; (18.16) and (18.21) involve nothing else. ■

The operator form of the duality, and the XY chain, are in the next reading.

## 18.10 The XY chain, and duality as a map

The TFIM is one member of a family,

$$H_{XY} = -J \sum_{n=1}^N \left( \frac{1+\eta}{2} \sigma_n^x \sigma_{n+1}^x + \frac{1-\eta}{2} \sigma_n^y \sigma_{n+1}^y \right) - h \sum_{n=1}^N \sigma_n^z, \quad (18.22)$$

with anisotropy  $\eta$ ;  $\eta = 1$  is the TFIM. (The literature writes  $\gamma$  for  $\eta$ ; this course keeps  $\gamma$  for [Lecture 16](#)'s coupling.) The XY chain goes through the same three steps as the TFIM, with the same sectors and grids, and gives [Theorem 18.6](#) with

$$\varepsilon(k) = 2\sqrt{(h - J \cos k)^2 + \eta^2 J^2 \sin^2 k}, \quad (18.23)$$

and signed  $e_0, e_\pi$  unchanged (checked against exact diagonalisation at  $N = 8$  by the check script; the derivation is an exercise). The gap closes where both squares vanish: on the line  $h = J$ , and on the segment  $\eta = 0, h \leq J$ , where  $\varepsilon$  vanishes at  $\cos k = h/J$  ([Figure 18.6](#)). At  $\eta = 0$  the chain is the XX chain, which conserves the number of fermions; its momentum grids are compared below.

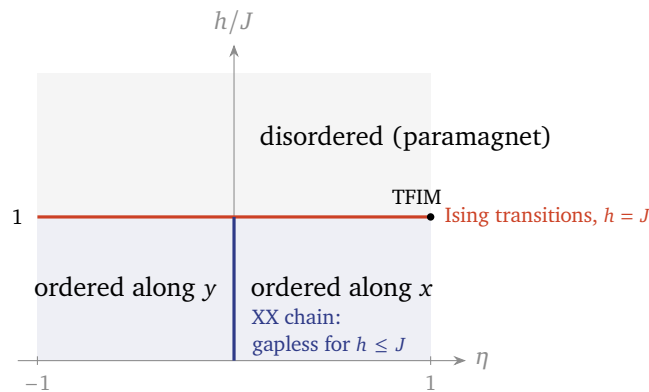


Figure 18.6: The XY chain's phase diagram in the plane of anisotropy  $\eta$  and field  $h/J$ . The gap closes on the line  $h = J$  and on the segment  $\eta = 0, h \leq J$ ; elsewhere it is open. The TFIM is the edge  $\eta = 1$ ; the transverse-field transition of this lecture is the dot. (A finer line inside the ordered phases, where correlations change their manner of decay, is omitted.)

*The XX chain* ( $\eta = 0$ ) conserves the fermion number, and its  $M$ -fermion eigenstates are Slater determinants of plane waves. As spin flips, hard-core bosons, two of them pick up only the exchange sign,  $S = -1$ , whatever their momenta, and [Lecture 10](#)'s periodicity condition becomes  $e^{ik_j N} = (-1)^{M-1}$ : exactly the NS or R grid of [\(18.10\)](#) for the sector of parity  $(-1)^M$ . The Heisenberg chain differs from the XX chain exactly in that its  $S$  depends on the rapidities.

*Duality.* Behind this is an exact map of operators (Kramers and Wannier 1941, for the classical two-dimensional Ising model) · *cited*: the domain-wall variables  $\tau^z = \sigma_n^x \sigma_{n+1}^x$  living on the bonds turn the bond term into a field and the field into a bond term, exchanging the ordered and disordered phases. It does not map the odd sector to itself — the signed mode  $e_0 = 2(h - J)$  changes sign — which is why the proposition says “even”. Kramers and Wannier's use of it survives intact: if there is one transition, it must be at the self-dual point.

Reading

## 18.11 The conserved charges

The TFIM's conserved charges are constructed directly, without a transfer matrix. Every occupation  $\gamma_k^\dagger \gamma_k$  commutes with  $H_\pm$ , but a single one is not local. Local combinations exist: weighting the term of  $k$  in (18.13) by  $\cos mk$  gives an operator that still commutes with  $H_\pm$ , block by block, and whose coefficients are still trigonometric polynomials in  $k$  — a sum of fermion bilinears of range at most  $m + 2$ . The first beyond  $H$  is, in spins,

$$Q_+ = \sum_{n=1}^N \left[ J(\sigma_n^x \sigma_{n+1}^z \sigma_{n+2}^x - \sigma_n^z) - h(\sigma_n^x \sigma_{n+1}^x + \sigma_n^y \sigma_{n+1}^y) \right], \quad [H, Q_+] = 0, \quad (18.24)$$

a local charge of range 3 (the tutorial, and checked exactly for  $N = 6, 8, 10$ ). Continuing in  $m$  gives about  $N$  local charges, as for XXX in Lecture 15's tower; Lecture 19 uses them to say which combinations of the  $n_k$  a relaxed state remembers.

Dictionary

the linear action–angle system,  $K(I) = \sum_i \nu_i I_i$ : independent oscillators, each angle turning at a fixed frequency, and nothing scatters (§ 6.3)  $\rightarrow$  free fermions,  $H = \sum_k \varepsilon(k)(n_k - \frac{1}{2})$ : independent modes,  $\gamma_k(t) = e^{-i\varepsilon(k)t} \gamma_k$ , two-body S-matrix  $\pm 1$  whatever the momenta (type A, structural analogy: the same role, no  $\hbar \rightarrow 0$  limit taken)

Reading

## 18.12 Correlations as a Toeplitz determinant

Write  $A_n = c_n^\dagger + c_n$  and  $B_n = c_n^\dagger - c_n$ , so that  $A_n^2 = 1$ ,  $B_n^2 = -1$ , and a bulk bond is  $\sigma_n^x \sigma_{n+1}^x = B_n A_{n+1}$  by (18.7). Since  $(\sigma^x)^2 = 1$ ,

$$\sigma_1^x \sigma_{1+r}^x = (\sigma_1^x \sigma_2^x)(\sigma_2^x \sigma_3^x) \cdots (\sigma_r^x \sigma_{r+1}^x) = B_1 A_2 B_2 A_3 \cdots B_r A_{r+1},$$

a product of  $2r$  fermion operators. In a Bogoliubov vacuum (a Gaussian state) such an expectation value is given by Wick's theorem as a Pfaffian of two-point functions, and because  $\langle A_l A_m \rangle = \delta_{lm}$  and  $\langle B_l B_m \rangle = -\delta_{lm}$  it collapses to a determinant (Lieb, Schultz and Mattis 1961) · *cited*:

$$\langle \sigma_1^x \sigma_{1+r}^x \rangle = \det [g(m-l+1)]_{l,m=1}^r, \quad g(d) = \langle B_n A_{n+d} \rangle = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-ikd} \frac{J e^{ik} - h}{|J e^{ik} - h|} dk$$

in the thermodynamic limit (for finite  $N$  the integral is the average over the NS grid). A matrix whose entries depend only on  $m - l$  is a *Toeplitz* matrix, and the large- $r$  behaviour of Toeplitz determinants is a classical subject (Szegő's theorem and its extensions). The results, due to Pfeuty (1970) and, for the XY chain, Barouch and McCoy (1971) · *cited*: for  $h < J$  the correlation tends to  $(1 - h^2/J^2)^{1/4}$ , so the spontaneous magnetisation is  $\langle \sigma^x \rangle = (1 - h^2/J^2)^{1/8}$ ; at  $h = J$  it decays like  $r^{-1/4}$ ; for  $h > J$  it decays exponentially. The check script confirms the determinant formula against exact diagonalisation at  $N = 12$ , and the ordered-phase limit at  $r = 200$  to six digits. Franchini's Chapter 1 and its appendices carry this out in full.

Reading

### 18.13 The open chain and Majorana edge modes

Cut the ring: drop the boundary bond. There is then no parity-dependent term, no sectors, and one quadratic Hamiltonian on all of Fock space. Rewrite each fermion as two Hermitian *Majorana* operators,  $\alpha_n = c_n + c_n^\dagger$  and  $\beta_n = i(c_n - c_n^\dagger)$ , with  $\alpha_n^2 = \beta_n^2 = 1$  and all different ones anticommuting. By (18.7) and  $1 - 2n_n = i\alpha_n\beta_n$ ,

$$H_{\text{open}} = -iJ \sum_{n=1}^{N-1} \beta_n \alpha_{n+1} - ih \sum_{n=1}^N \alpha_n \beta_n.$$

Each fermion is split in two, the field pairs the two halves on one site and the bond pairs halves on neighbouring sites. At  $h = 0$  the bonds pair  $\beta_n$  with  $\alpha_{n+1}$ , and two Majoranas are left over,  $\alpha_1$  at the left end and  $\beta_N$  at the right, appearing nowhere in  $H$ . The fermion  $f = \frac{1}{2}(\alpha_1 + i\beta_N)$  commutes with  $H$ : it costs zero energy, every level is exactly doubly degenerate, and the two halves of this zero mode sit at opposite ends of the chain. In spin language these are, again, the two ferromagnets. For  $0 < h < J$  the zero mode survives with an energy exponentially small in  $N$  (the check script measures  $(h/J)^N$ ), localised near the ends; for  $h > J$  it is gone. This is Kitaev's chain (2001) · *cited*, a model of a one-dimensional  $p$ -wave superconductor: in the fermion language the ground-state doublet stays split only exponentially under *sufficiently weak* local perturbations that conserve fermion parity and do not close the bulk gap (a physical perturbation of fermions always conserves parity; the exact pairing of every level at  $h = 0$  is not what is protected), while in the spin language a weak longitudinal field  $\sigma^x$ , which is local for spins, splits it at once. The same algebra, read in two languages, gives a fragile symmetry breaking and a robust topological degeneracy (Franchini, §1.5).

Reading

### 18.14 Impenetrable bosons are free fermions

This lecture solved a chain by turning it into free fermions. [Lecture 16](#)'s interacting Bose gas, pushed to infinite repulsion, *becomes* free fermions — exactly, for every state and not only the ground state, by Girardeau's map  $\psi_B = \prod_{i < j} \text{sgn}(x_i - x_j) \psi_F$  · *cited: Girardeau (1960)*. On a *ring* the map carries a boundary condition: taking one coordinate once round reverses its order with the other  $N - 1$ , so periodic bosons need fermions periodic for odd  $N$  and *antiperiodic* for even  $N$ . These are continuum fermions,  $\epsilon = k^2$ ; this lecture's live on a lattice with its own dispersion. What the two share is the structure, not the Hamiltonian.

#### Summary.

- *The model* (18.1) and its parity  $\Pi$ , which splits the chain into an even and an odd sector.
- *Jordan–Wigner* ([Theorem 18.3](#)): spins become fermions through a string; the Hamiltonian is quadratic in each sector, with antiperiodic (even) or periodic (odd) boundary conditions.
- *Fourier and Bogoliubov* ([Theorem 18.6](#)): two grids of momenta, quasiparticles of energy  $\epsilon(k)$ , and every eigenvalue, sector by sector.

- Next: [Lecture 19](#) quenches the chain and asks what it relaxes to.

**Tutorial.** Exercise Sheet 18.

## References

P. Jordan and E. Wigner, Z. Phys. 47 (1928) 631 (the map); E. Lieb, T. Schultz and D. Mattis, Ann. Phys. 16 (1961) 407 (free fermions, the Toeplitz determinant); P. Pfeuty, Ann. Phys. 57 (1970) 79 (the TFIM solved); F. Franchini, *An Introduction to Integrable Techniques for One-Dimensional Quantum Systems*, Springer (2017), Chapter 1 (the XY chain, in full); G. B. Mbeng, A. Russomanno and G. E. Santoro, *The quantum Ising chain for beginners*, arXiv:2009.09208 (the two parity sectors, carefully); H. A. Kramers and G. H. Wannier, Phys. Rev. 60 (1941) 252; A. Yu. Kitaev, Phys.-Usp. 44 (2001) 131; M. Girardeau, J. Math. Phys. 1 (1960) 516.